

Lecture 4: Classification models

TECH6

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See GitHub repository for notebooks and data:

<https://github.com/richardfoltyn/TECH6-H26>

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1 Classification models with Scikit-learn

1.1 Logistic regression

In the previous lecture, we studied how to predict continuous variables using linear models with shrinkage. We now turn to one of the most classical machine learning tasks, classification. To this end, we study the simplified setting where we have only two classes (labeled 0 and 1) so that the outcome is $y_i \in \{0, 1\}$, and we want to train a model to predict the class membership of observations based on some features $\mathbf{x}_i$.

The classical approach to perform this task is to assume that the probability of the outcome 1 is given by

$$p(\mathbf{x}_i) \equiv \text{Prob}(y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\mu + \mathbf{x}_i' \boldsymbol{\beta}))}$$

where the function on the right $\sigma(z) = \frac{1}{1+e^{-z}}$ is called the **sigmoid function** in machine learning or the **logistic** cumulative distribution function (CDF) in statistics and econometrics. Note that this function maps any (positive or negative) real value z into a value between $(0, 1)$, and hence is well-suited to model the probability that $y_i = 1$.

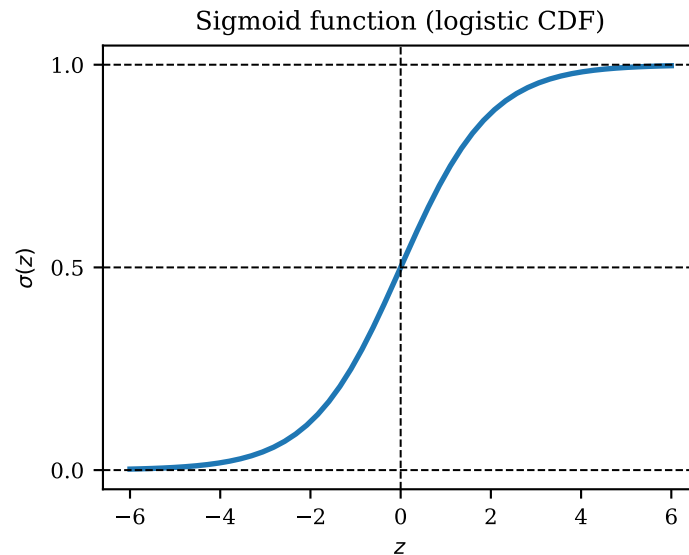
We can easily verify this by plotting the logistic CDF from `scipy.stats`:

```
[1]: import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import logistic

zvalues = np.linspace(-6, 6, 50)

plt.plot(zvalues, logistic.cdf(zvalues), lw=2.0)
```

```
# Add horizontal and vertical lines
for y in (0.0, 0.5, 1.0):
    plt.axhline(y, ls='--', lw=0.75, c='black')
plt.axvline(0.0, ls='--', lw=0.75, c='black')
plt.xlabel('$z$')
plt.ylabel(r'$\sigma(z)$')
plt.yticks([0.0, 0.5, 1.0])
_ = plt.title('Sigmoid function (logistic CDF)')
```



As you can see, the value of $\sigma(z)$ tends to 0 for small (negative) z , and it approaches 1 for large values of z .

In machine learning, logistic regression is categorized as a linear model (see the scikit-learn user guide on [linear models](#)) even though $\sigma(z)$ is clearly not linear. This is because the index z_i that enters the sigmoid function is assumed to be linear in the features $\mathbf{x}_i$,

$$z_i = \mu + \mathbf{x}_i' \boldsymbol{\beta}$$

Consequently, one can show that the log odds are a linear function of features:

$$\log \left(\frac{p(\mathbf{x}_i)}{1 - p(\mathbf{x}_i)} \right) = \mu + \mathbf{x}_i' \boldsymbol{\beta}$$

Before we can fit the model using logistic regression, we need to define an appropriate loss function. This loss function is derived from the log-likelihood function of a binary logit model, defined as

$$\mathcal{L}(\mu, \boldsymbol{\beta}) = \sum_{i=1}^N y_i \log(p(\mathbf{x}_i)) + (1 - y_i) \log(1 - p(\mathbf{x}_i))$$

The details underlying the derivation of this likelihood function are beyond the scope of this course. For our purposes, it's sufficient to know that in maximum likelihood estimation (MLE), the parameters μ and $\boldsymbol{\beta}$ are chosen so that the log-likelihood $\mathcal{L}$ of observing a given sample of $(y_i, \mathbf{x}_i)$ is *maximized*.

To derive the loss function L for the logistic regression, we simply take the *negative* log-likelihood function and find the parameters $(\mu, \boldsymbol{\beta})$ which *minimize* the loss L :

$$L(\mu, \boldsymbol{\beta}) = - \underbrace{\frac{1}{N} \mathcal{L}(\mu, \boldsymbol{\beta})}_{\text{scaled log-likelihood}} + \underbrace{\frac{r(\boldsymbol{\beta})}{C}}_{\text{regularization}}$$

Moreover, we have already incorporated the scaled regularization term $\frac{r(\beta)}{C}$, where $r(\bullet)$ is a function of β , which is used analogously to Ridge regression and Lasso to shrink the parameters β and prevent overfitting. The fitted logistic regression parameters are thus

$$(\hat{\mu}, \hat{\beta}) = \arg \min_{\mu, \beta} L(\mu, \beta)$$

1.1.1 Example: Predicting binary class membership

To illustrate how to work with logistic regression, we create an artificial sample with two features (x_{1i}, x_{2i}) that are mapped into a binary outcome $y_i \in \{0, 1\}$. We restrict our attention to only two features since this makes it easy to plot decision regions.

Creating a demo dataset

Assume that the data-generating process can be characterized as follows:

$$y_i = \begin{cases} 1 & \text{if } f(x_{1i}, x_{2i}) + \epsilon_i \geq 0 \\ 0 & \text{else} \end{cases}$$

where $f(\bullet)$ is some deterministic function of (x_{1i}, x_{2i}) and ϵ_i is an error term. For illustration, we'll again assume that $f(\bullet)$ is a combination of trigonometric functions which we'll attempt to approximate using polynomials. Specifically, the model studied in this section is

$$f(x_{1i}, x_{2i}) = \sin(2\pi x_{1i}) \cos(\pi x_{2i})$$

$$\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2)$$

The code below defines $f(x_{1i}, x_{2i})$ and plots the function on a rectangular grid which we create with `np.meshgrid()`.

```
[2]: import numpy as np
```

```
def f(x1, x2):
    """
    True function for classification examples
    """
    return np.sin(2 * np.pi * x1) * np.cos(np.pi * x2)
```

```
[3]: import matplotlib.pyplot as plt
```

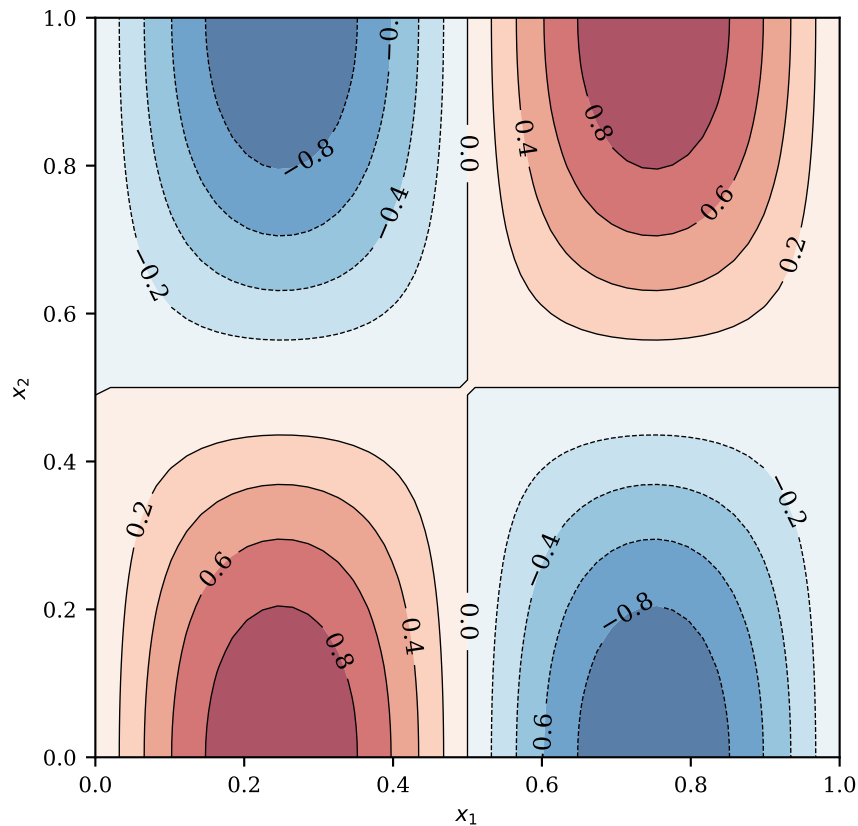
```
# x-values for grid
x = np.linspace(0, 1, 50)

# Create coordinates for x1 and x2
x1_grid, x2_grid = np.meshgrid(x, x)

# Deterministic function f(x1, x2)
z = f(x1_grid, x2_grid)

# Contour plot of deterministic function
plt.figure(figsize=(5, 5))
levels = np.linspace(-1, 1, 11)
cntrf = plt.contourf(x1_grid, x2_grid, z, levels=levels, cmap='RdBu_r', alpha=0.7)
CS = plt.contour(x1_grid, x2_grid, z, levels=levels, linewidths=0.5, colors='k')
plt.clabel(CS, CS.levels, fontsize=9)
```

```
plt.xlabel('$x_1$')
_ = plt.ylabel('$x_2$')
```



As you see, this function takes on positive values in two quadrants (where y_i is thus 1) and negative values in the other two (where y_i is therefore 0). Without some noise ϵ_i , class membership would be trivial to predict, so we next create a sample that incorporates noise.

To do this, we define the function `create_class_data()` which we'll use throughout the lecture to create a demo dataset for classification of the desired size:

```
[4]: def create_class_data(N=1000, sigma=0.2, rng=None):
    """
    Create synthetic data for binary classification examples.

    Parameters
    -----
    N : int
        Number of observations to generate.
    sigma : float
        Standard deviation of the error term.
    rng : numpy.random.Generator, optional
        Random number generator to use.

    Returns
    -----
    X : array-like
        Feature matrix.
    y : array-like
        Binary response variable.

    """
    if rng is None:
```

```

    rng = np.random.default_rng(seed=1234)

    # Draw features from uniform distribution
    x1 = rng.uniform(0, 1, size=N)
    x2 = rng.uniform(0, 1, size=N)

    z = f(x1, x2)

    # Add noise to latent variable if noise variance is positive
    if sigma > 0:
        # Draw errors from normal distribution
        epsilon = rng.normal(0, sigma, size=N)
        z += epsilon

    # Convert latent variable to binary response variable
    y = (z >= 0).astype(int)

    # Stack features into matrix
    X = np.column_stack((x1, x2))

    return X, y

```

We can now use this function to create the demo dataset:

```

[5]: # Sample size
    N = 100

    # Standard deviation of noise
    sigma_eps = 0.2

    # Create demo dataset for classification
    X, y = create_class_data(N=N, sigma=sigma_eps)

```

We can verify that y_i takes on only two values with class labels 0 and 1:

```

[6]: classes = np.unique(y)
    classes

```

```

[6]: array([0, 1])

```

Next, we import the function `plot_classes()` defined in `lecture4_classifiers.py` which creates a scatter plot from the sample and uses different marker symbols based on the class labels.

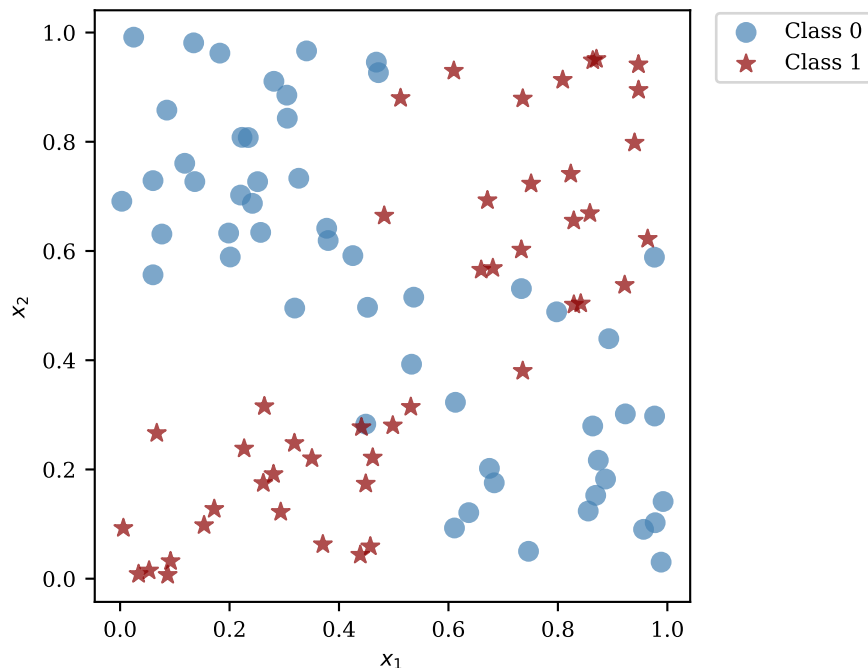
We will use this function repeatedly to plot the sample data and model predictions. The figure below visualizes the artificial sample we have just created. As you can see, unlike in the previous graph, the class membership is no longer clearly split between quadrants due to the additional noise ϵ_i .

```

[7]: from lecture4_classifiers import plot_classes

    # Plot demo dataset for classification
    _ = plot_classes(X, y)

```



1.1.2 Fitting a simple model

As usual, before fitting the model, we first split the sample into a training and a test set. Unlike with linear regression models (Ridge, Lasso, OLS), it is advantageous to use *stratification* when splitting the sample so that both the training and test samples contain an approximately equal share of each class. We accomplish this by specifying the `stratify` keyword argument which passes the values of the target variable y for stratification.

```
[8]: from sklearn.model_selection import train_test_split

# Split data into training and test sets, stratifying by class
X_train, X_test, y_train, y_test = train_test_split(
    X, y, stratify=y, test_size=0.3, random_state=1234
)
```

Of course, given randomness and the small sample size used here, the share of classes will not be exactly identical across the training and test samples, as the following code shows:

```
[9]: import pandas as pd

print('Distribution of classes in training sample:')
print(pd.Series(y_train).value_counts() / len(y_train))

print('\nDistribution of classes in test sample:')
print(pd.Series(y_test).value_counts() / len(y_test))
```

Distribution of classes in training sample:

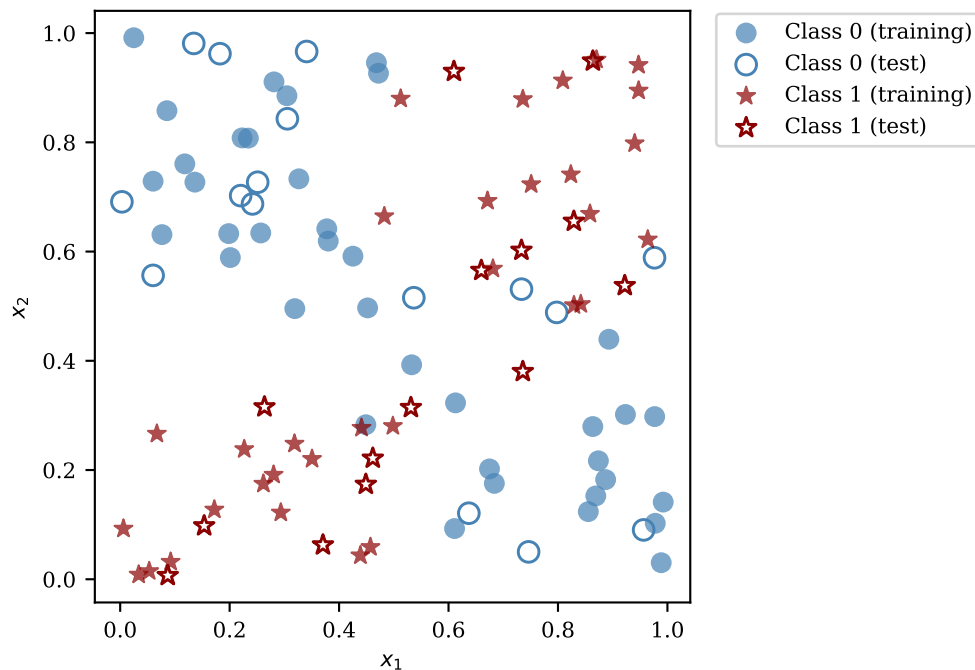
```
0    0.542857
1    0.457143
Name: count, dtype: float64
```

Distribution of classes in test sample:

```
0    0.533333
1    0.466667
Name: count, dtype: float64
```

We can now plot the training and test samples, using different marker symbols for each:

```
[10]: _ = plot_classes(X_train, y_train, X_test, y_test)
```



Specifying the regularization strength

We proceed to fit the model using the `LogisticRegression` estimator from `scikit-learn`. `LogisticRegression` implements four different types of regularization which correspond to different functional forms for $r(\bullet)$ in the penalty term:

1. the *L1 penalty* we encountered with Lasso;
2. the *L2 penalty* from the Ridge regression;
3. the convex combination of these used for the *elastic net*; or
4. *no penalty* (this then corresponds to the classical binary logit used in econometrics to model discrete choice).

Unfortunately, the way to specify the desired type of regularization changed in `scikit-learn` version 1.8, which is the version in the course environment. While the previous arguments are still supported, they will produce warnings and will be completely removed in version 1.10. You should therefore avoid using them, if possible, and only specify the arguments `l1_ratio` and `C`.

On the other hand, if you are on a Mac with an Intel CPU, your Anaconda environment will most likely have the outdated version 1.6. In that case, you have to use the old way of specifying the penalty using the `penalty` argument. The code sections below contain comments to point out how functions need to be called with `scikit-learn` prior to version 1.8.

You can check which version you are running as follows:

```
[11]: import sklearn
```

```
sklearn.__version__
```

```
[11]: '1.9.0'
```

The following table lists the available regularization terms and their corresponding `l1_ratio` (version ≥ 1.8) and `penalty` (version < 1.8) arguments that can be used when creating an instance of

LogisticRegression. Note that the argument C is the *inverse* penalty, i.e., *higher* values of C *decrease* the strength of the regularization term.

Penalty type	$r(\beta)$	penalty (version < 1.8)	l1_ratio (version $\geq$ 1.8)	C
L1	$r(\beta) = \ \beta\ _1 = \sum_{k=1}^K \ \beta_k\ $	'l1'	1	desired inverse penalty
L2	$r(\beta) = \frac{1}{2} \ \beta\ _2^2 = \frac{1}{2} \sum_{k=1}^K \beta_k^2$	'l2'	0	desired inverse penalty
L1 and L2	$r(\beta) = \rho \ \beta\ _1 + \frac{1-\rho}{2} \ \beta\ _2^2$	'elasticnet'	(0,1)	desired inverse penalty
None	—	None	—	np.inf

Fitting the logistic regression model

We first experiment with the simplest possible linear model of two explanatory variables where

$$z_i = \mu + \beta_1 x_{1i} + \beta_2 x_{2i}$$

so that the probability of observing class 1 is

$$p(x_i) \equiv \text{Prob}(y_i = 1 \mid x_i) = \frac{1}{1 + \exp(-(\mu + \beta_1 x_{1i} + \beta_2 x_{2i}))}$$

For now we skip including a regularization term by specifying C = np.inf for scikit-learn version 1.8+, or penalty=None for versions prior to 1.8.

```
[12]: from sklearn.linear_model import LogisticRegression

# Create logistic regression model without regularization penalty
lr = LogisticRegression(C=np.inf, fit_intercept=True, max_iter=1000)

# NOTE: For sklearn version < 1.8, use penalty=None instead of C=np.inf
# lr = LogisticRegression(penalty=None, fit_intercept=True, max_iter=1000)
```

```
[13]: lr.fit(X_train, y_train)
```

```
[13]: LogisticRegression(C=inf, max_iter=1000)
```

Note that in scikit-learn 1.8.0, a warning is generated due to a bug in the code. However, the model estimation without a penalty still works as intended.

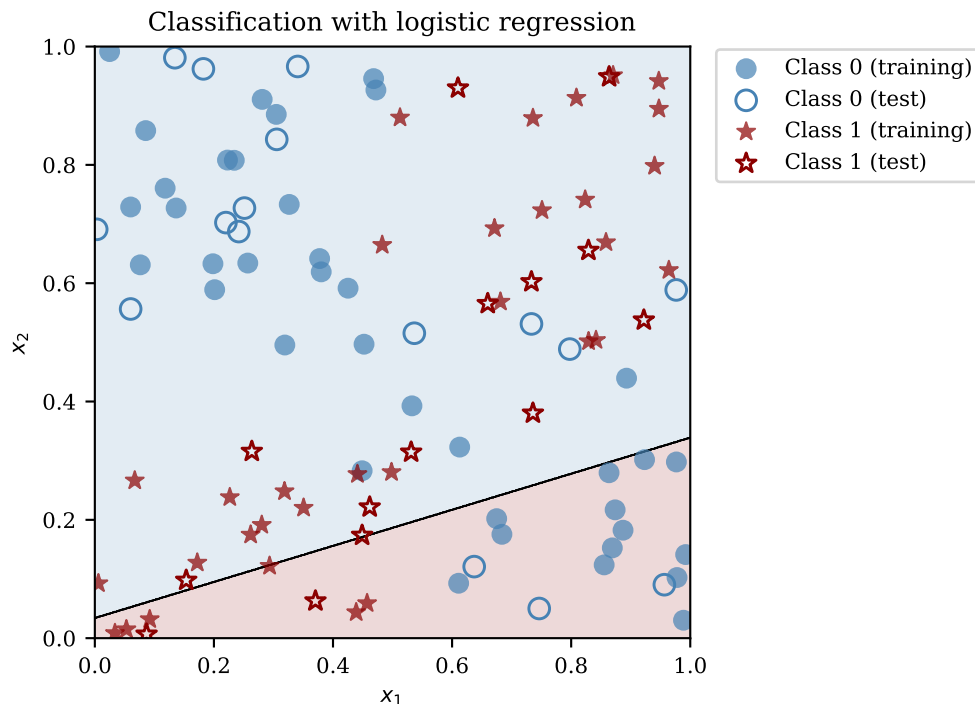
1.1.3 Assessing model accuracy

We first plot the decision boundaries implied by the fitted model. To this end, we use the helper function plot_decision_boundary() implemented in lecture4_classifiers.py. The function takes as argument an existing Matplotlib Axes object and adds the decision boundaries to an existing figure.

```
[14]: from lecture4_classifiers import plot_decision_boundary

# Create x-values to evaluate decisions (used for either dimension)
xvalues = np.linspace(0, 1, 1000)

ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, lr)
_ = ax.set_title('Classification with logistic regression')
```



As you see, the simple linear model does a poor job predicting class membership and misclassifies many observations. The graph shows that for any combination (x_{1i}, x_{2i}) falling into the blue area, the fitted model predicts $y_i = 0$, whereas in the red area it predicts $y_i = 1$.

We can explicitly compute the accuracy, i.e., the share of correctly predicted class labels, using the `score()` method of the estimator, or the `accuracy_score()` function. These return the same results.

```
[15]: # Predict y on training and test samples
y_train_pred = lr.predict(X_train)
y_test_pred = lr.predict(X_test)

[16]: from sklearn.metrics import accuracy_score

# Compute accuracy on training and test sets
acc_train = accuracy_score(y_train, y_train_pred)
acc_test = accuracy_score(y_test, y_test_pred)

print(f'Accuracy on training sample: {acc_train:.3f}')
print(f'Accuracy on test sample: {acc_test:.3f}')
```

```
Accuracy on training sample: 0.443
Accuracy on test sample: 0.500
```

Alternatively, we can use the `score()` method directly to compute the same statistic, for example on the test sample:

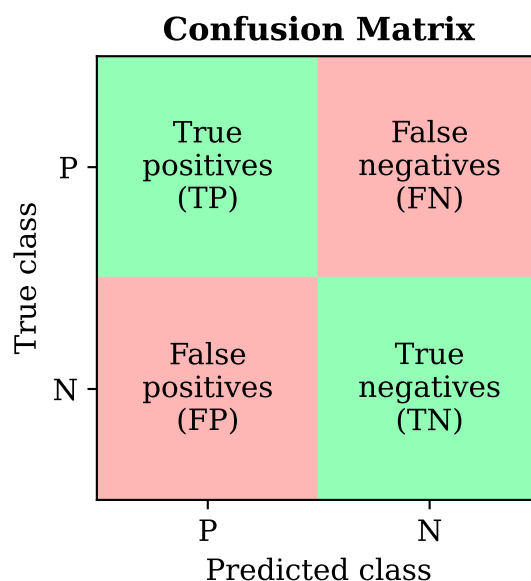
```
[17]: acc_test = lr.score(X_test, y_test)
print(f'Accuracy on test sample: {acc_test:.3f}')
```

```
Accuracy on test sample: 0.500
```

There are other metrics to assess how well the model predicts class labels. These usually combine the elements of the following confusion matrix in some way.

First, consider the confusion matrix for a generic classification problem (with labels P and N for Positive and Negative), plotted below:

```
[18]: from lecture4_classifiers import plot_generic_confusion_matrix
      plot_generic_confusion_matrix()
```



As you see, a classifier would ideally make predictions on some sample of known labels which are located in the green quadrants. Using the labels of this matrix, the accuracy score we have computed above is given by

$$ACC = \frac{TP + TN}{FP + FN + TP + TN}$$

However, we might be interested in different measures, for example *precision*, defined as

$$PRE = \frac{TP}{TP + FP}$$

which is the share of true positives among all observations that are predicted as positive by the model, or *recall*, defined as

$$REC = \frac{TP}{FN + TP}$$

which is the share of true positives among all observations that are truly positive in the data.

Another commonly used scoring metric is the so-called *F1 score*, computed as

$$F1 = 2 \frac{PRE \cdot REC}{PRE + REC}$$

Which of these metrics to use depends on the scenario. For example, we could have a sample where there are only very few positive observations, so a model that always predicts the negative label will have a high accuracy. On the other hand, it will do poorly when measured by *recall* because its *TP* count would be zero.

scikit-learn implements many commonly used scoring metrics in the module `sklearn.metrics`. Using functions from that module, we can compute the metrics defined above:

```
[19]: from sklearn.metrics import f1_score, precision_score, recall_score

      # Accuracy (same as above)
      acc_test = accuracy_score(y_test, y_test_pred)
      # Precision
      pre = precision_score(y_test, y_test_pred)
      # Recall
      rec = recall_score(y_test, y_test_pred)
```

```
# F1 score
f1 = f1_score(y_test, y_test_pred)

print(f'Accuracy: {acc_test:.3f}')
print(f'Precision: {pre:.3f}')
print(f'Recall: {rec:.3f}')
print(f'F1 score: {f1:.3f}')
```

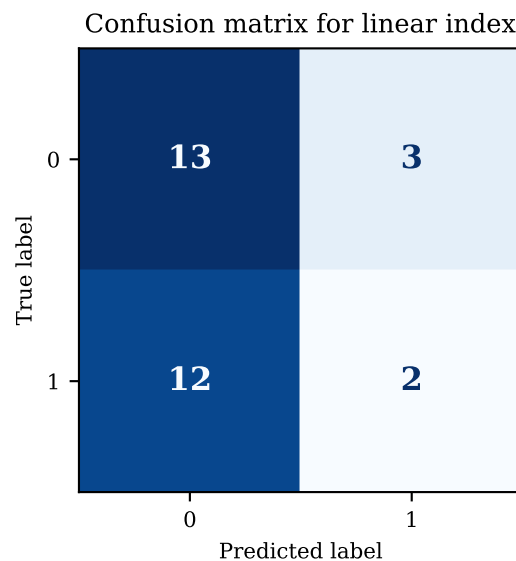
Accuracy: 0.500
Precision: 0.400
Recall: 0.143
F1 score: 0.211

As you can see, the logistic regression with a linear index performs even more poorly on some of these metrics than suggested by the accuracy score.

Finally, we can plot the confusion matrix using the class `ConfusionMatrixDisplay`. This plot can be created from the predicted values or directly from the estimator. We demonstrate how the confusion matrix can be plotted from the predicted values below:

```
[20]: from sklearn.metrics import ConfusionMatrixDisplay

# Plot confusion matrix from predicted values
_ = ConfusionMatrixDisplay.from_predictions(
    y_test,
    y_test_pred,
    colorbar=False,
    cmap='Blues',
    text_kw={'fontsize': 12, 'fontweight': 'bold'},
).ax_.set_title('Confusion matrix for linear index')
```



1.1.4 Digression: linear decision boundary

You might wonder why the decision boundary plotted earlier is a straight line even though the sigmoid function is clearly nonlinear. This arises because LogisticRegression assigns predictions based on the most likely class as determined by the predicted probabilities of belonging to each class for each observation. We can predict these underlying probabilities using the `predict_proba()` method, for example for the test sample:

```
[21]: # Predict probabilities for first five test samples
lr.predict_proba(X_test)[:5]
```

```
[21]: array([[0.51830556, 0.48169444],
          [0.55159038, 0.44840962],
          [0.59726736, 0.40273264],
          [0.46472375, 0.53527625],
          [0.57511958, 0.42488042]])
```

Based on these predicted probabilities, LogisticRegression assigns the predicted class as the most likely outcome. With only two categories, $y_i = 1$ is therefore predicted whenever the value in the second column is larger than 0.5. Plugging this into the sigmoid function, we have

$$\begin{aligned}
 y_i = 1 &\iff \sigma(\mu + \beta_1 x_{1i} + \beta_2 x_{2i}) \geq 0.5 \\
 &\iff \frac{1}{1 + \exp(-\mu - \beta_1 x_{1i} - \beta_2 x_{2i})} \geq 0.5 \\
 &\iff \exp(-\mu - \beta_1 x_{1i} - \beta_2 x_{2i}) \leq 1 \\
 &\iff \mu + \beta_1 x_{1i} + \beta_2 x_{2i} \geq 0 \\
 &\iff x_{2i} \leq -\frac{\mu}{\beta_2} - \frac{\beta_1}{\beta_2} x_{1i}
 \end{aligned}$$

The last step above follows because $\beta_2 < 0$ in this example as you can verify by checking the attribute `lr.coef_`. The take-away from these derivations is that any combination of features x_{1i}, x_{2i} is mapped to $y_i = 1$ if x_{2i} lies underneath the line defined by $-\frac{\mu}{\beta_2} - \frac{\beta_1}{\beta_2} x_{1i}$, and therefore the decision boundary is a *straight line*.

1.1.5 Fitting a model with polynomials

We now extend the simple model and approximate the true trigonometric relationship between y_i and x_i using polynomials in x_{1i}, x_{2i} where we allow for interaction terms up to some upper bound:

$$z_i = \mu + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \beta_4 x_{1i}^2 + \beta_5 x_{2i}^2 + \dots$$

```
[22]: from sklearn.linear_model import LogisticRegression
      from sklearn.pipeline import make_pipeline
      from sklearn.preprocessing import PolynomialFeatures, StandardScaler

      # Maximum polynomial degree
      degree = 5

      # Create logistic regression model without regularization penalty
      lr = LogisticRegression(fit_intercept=True, C=np.inf, max_iter=1000, random_state=1234)

      # NOTE: For sklearn version < 1.8, use penalty=None instead of C=np.inf
      # lr = LogisticRegression(
      #     fit_intercept=True, penalty=None, max_iter=1000, random_state=1234
      # )

      # Create pipeline with polynomial features and logistic regression
      pipe_lr = make_pipeline(
          PolynomialFeatures(degree=degree, include_bias=False), StandardScaler(), lr
      )
```

At this point we don't strictly need to scale the polynomial features using the `StandardScaler`, but this will be necessary below when we introduce a regularization term.

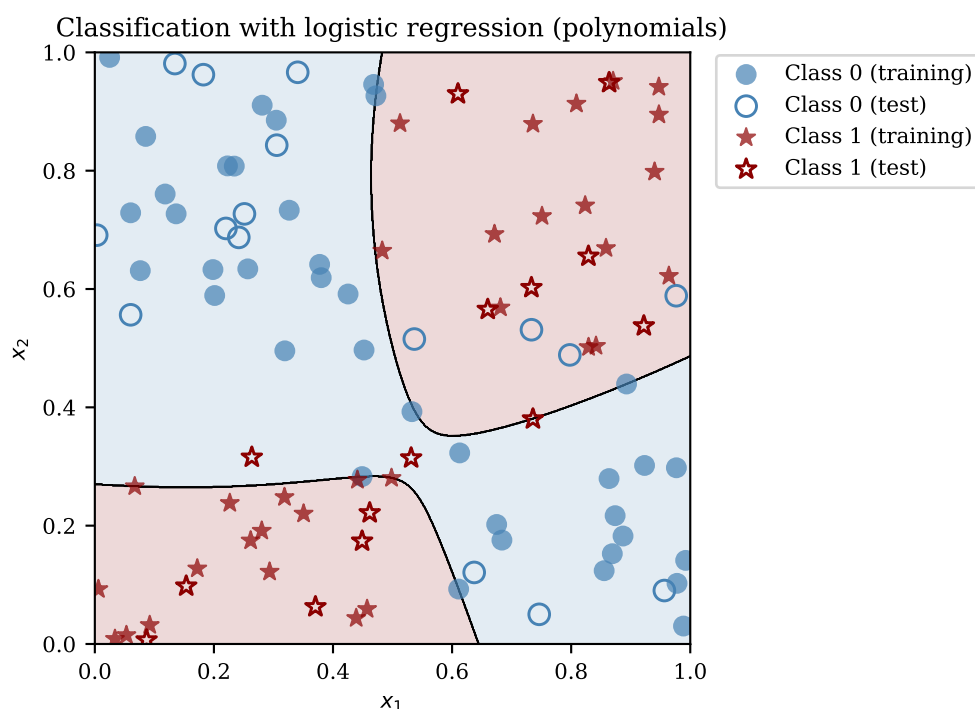
We can now fit the model on the training data:

```
[23]: pipe_lr.fit(X_train, y_train)
```

```
[23]: Pipeline(steps=[('polynomialfeatures',
                        PolynomialFeatures(degree=5, include_bias=False)),
                      ('standardscaler', StandardScaler()),
                      ('logisticregression',
                        LogisticRegression(C=inf, max_iter=1000, random_state=1234))])
```

Using the functions defined above, we can plot the decision boundaries of this more complex model:

```
[24]: # Create x-values used to evaluate decisions
xvalues = np.linspace(0, 1, 1000)
ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, pipe_lr)
_ = ax.set_title('Classification with logistic regression (polynomials)')
```



The substantial increase in model fit is immediately apparent. Using the accuracy score, we can confirm that the model is indeed much better at predicting the class labels on the test sample:

```
[25]: # Predict y on test sample
y_test_pred = pipe_lr.predict(X_test)

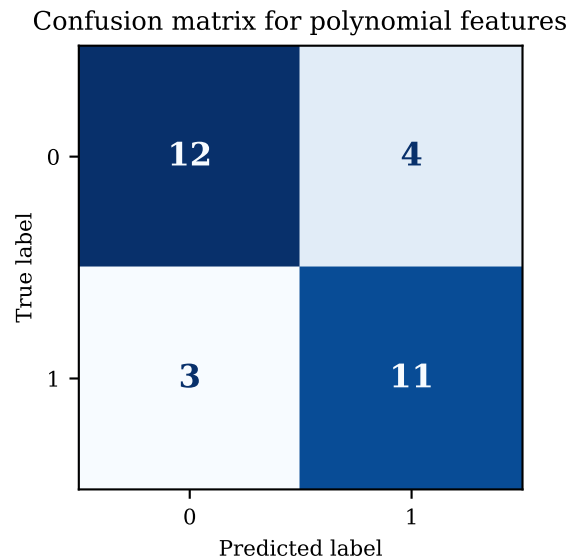
acc_test = accuracy_score(y_test, y_test_pred)
pre_test = precision_score(y_test, y_test_pred)
rec_test = recall_score(y_test, y_test_pred)

print(f'Accuracy on test sample: {acc_test:.3f}')
print(f'Precision on test sample: {pre_test:.3f}')
print(f'Recall on test sample: {rec_test:.3f}')
```

```
Accuracy on test sample: 0.767
Precision on test sample: 0.733
Recall on test sample: 0.786
```

Using polynomials, the model predicts around 77% of cases in the test sample correctly. Plotting the confusion matrix, we also see that more observations lie on the diagonals.

```
[26]: # Plot confusion matrix from predicted values
_ = ConfusionMatrixDisplay.from_predictions(
    y_test,
    y_test_pred,
    colorbar=False,
    cmap='Blues',
    text_kw={'fontsize': 12, 'fontweight': 'bold'},
).ax_.set_title('Confusion matrix for polynomial features')
```



1.1.6 Cross-validating the penalty term

So far, we haven't introduced a regularization term when fitting the logistic regression. Analogous to Ridge regression and Lasso, regularization can be used to prevent overfitting.

Since we don't know the "best" regularization parameter C , we use `LogisticRegressionCV` to perform cross-validation.

As is the case with most bespoke cross-validation classes in `scikit-learn`, `LogisticRegressionCV` does not support pipelines, so we need to manually create the polynomials and apply standardization on the training set before we run cross-validation:

```
[27]: from sklearn.linear_model import LogisticRegressionCV

# Pipeline to create polynomial features and standardize them
transform = make_pipeline(
    PolynomialFeatures(degree=degree, include_bias=False), StandardScaler()
)

# Fit and transform training data
X_train_poly = transform.fit_transform(X_train)
```

We can now run the cross-validation. For this we choose the L2 penalty (the same as for Ridge regression), we use 10 folds, and tell `LogisticRegressionCV` to choose 50 values of C which are uniformly spaced in logs between the values $1e-4$ and $1e4$.

```
[28]: # Create and run Logistic regression cross-validation
lrcv = LogisticRegressionCV(
    Cs=50, # Number of C to try, spaced uniformly in logs
    cv=10, # Number of folds
    fit_intercept=True, # Fit intercept
    l1_ratios=(0.0,), # Use L2 penalty
```

```

max_iter=1000, # Maximum iterations (increase from default)
n_jobs=4, # Number of parallel jobs
use_legacy_attributes=True, # Use legacy attributes in sklearn 1.8+
random_state=1234, # Random seed
)

# NOTE: For sklearn version < 1.8, use penalty='l2' instead of l1_ratios=(0.0, )
# lrcv = LogisticRegressionCV(
#     Cs=50, # Number of C to try, spaced uniformly in logs
#     cv=10, # Number of folds
#     fit_intercept=True, # Fit intercept
#     penalty='l2', # Use L2 penalty
#     max_iter=1000, # Maximum iterations (increase from default)
#     n_jobs=4, # Number of parallel jobs
#     random_state=1234, # Random seed
# )

```

The argument `use_legacy_attributes=True` specified above instructs scikit-learn to create attributes of the fitted object that are the same as in versions prior to 1.8. You can skip this argument if you want to follow the new attribute definitions in 1.8, but we'll continue to use the old variants in this lecture so the code works also for those using older versions of scikit-learn. This affects the attributes `C_`, `l1_ratio_`, `coefs_paths_`, `scores_`, and `n_iter_`. See the [release notes](#) for details if you plan to access any of these attributes after running the cross-validation.

```

[29]: # Run cross-validation
lrcv.fit(X_train_poly, y_train)

```

```

/home/richard/.conda/envs/TECH6/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:2137: FutureWarning: The default
value of the parameter 'scoring' will change from None, i.e., accuracy, to
'neg_log_loss' in version 1.11. To silence this warning, explicitly set the
scoring parameter: scoring='neg_log_loss' for the new, scoring='accuracy' or
scoring=None for the old default.
  warnings.warn(

```

```

[29]: LogisticRegressionCV(Cs=50, cv=10, l1_ratios=(0.0,), max_iter=1000, n_jobs=4,
random_state=1234, use_legacy_attributes=True)

```

Once the cross-validation is done, we can extract the best penalty parameter C from the array attribute `C_`.

In this case, the optimal C is given by

```

[30]: print(f'Best C: {lrcv.C_[0]:.4f}')

```

```
Best C: 35.5648
```

Note that unlike the penalty parameters used for Ridge and Lasso, C in this case is the *inverse* penalty strength, i.e., larger values of C introduce a *smaller* penalty.

Finally, we recreate the whole pipeline we had earlier but use the optimal penalty parameter. This is not necessary if we just want to compute predicted values or accuracy metrics, but the plotting functions we have created expect the estimator pipeline to also create the polynomial features and perform standardization.

```

[31]: # Logistic regression model with optimal C from cross-validation
lr_opt = LogisticRegression(
    C=lrcv.C_[0], fit_intercept=True, l1_ratio=0.0, max_iter=1000, random_state=1234
)

# NOTE: For sklearn version < 1.8, use penalty='l2' instead of l1_ratio=0.0
# lr_opt = LogisticRegression(
#     C=lrcv.C_[0], fit_intercept=True, penalty='l2', max_iter=1000, random_state=1234
# )

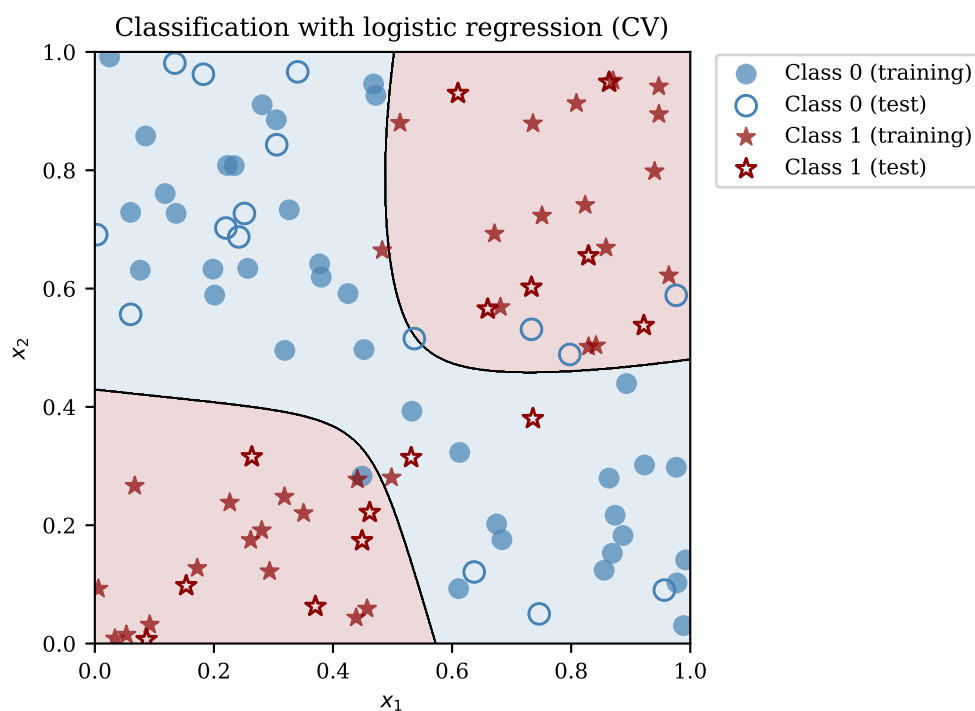
```

```
# )

# Create pipeline using cross-validated C
lr_pipe_opt = make_pipeline(
    PolynomialFeatures(degree=degree, include_bias=False),
    StandardScaler(),
    lr_opt,
).fit(X_train, y_train)
```

We reuse the plotting code from above to plot the decision boundaries for the optimal model:

```
[32]: # Create x-values used to evaluate decisions
xvalues = np.linspace(0, 1, 1000)
ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, lr_pipe_opt)
_ = ax.set_title('Classification with logistic regression (CV)')
```



There are slight improvements compared to the model without the penalty term. We can also confirm that the accuracy score for the test data increased from around 77% to 83%:

```
[33]: # Compute accuracy of cross-validated model on test data
y_test_pred = lr_pipe_opt.predict(X_test)

acc_test = accuracy_score(y_test, y_test_pred)
pre_test = precision_score(y_test, y_test_pred)
rec_test = recall_score(y_test, y_test_pred)

print(f'Accuracy on test sample: {acc_test:.3f}')
print(f'Precision on test sample: {pre_test:.3f}')
print(f'Recall on test sample: {rec_test:.3f}')
```

```
Accuracy on test sample: 0.833
Precision on test sample: 0.800
Recall on test sample: 0.857
```

Note that the model above has a second hyperparameter, the maximum number of polynomial degrees, which we fixed at 5 for the entire cross-validation. In theory, we could perform the cross-validation

jointly on the polynomial degree and C at the same time for even better results.

1.1.7 Validation curves

We conclude this section with plotting the validation curve for the above example as we vary the penalty parameter C. Recall that the validation curve plots some metric of model performance (in this case the accuracy score) computed from the validation sample against a range of values for a hyperparameter, in this case C (see the scikit-learn [documentation](#) for details).

We can leverage scikit-learn's `validation_curve()` function to perform this task for us. Since `validation_curve()` does work with pipelines, we first create the usual pipeline to create the polynomial features, perform standardization, and add a logistic regression estimator where we specify penalty to be of L2 type.

```
[34]: lr = LogisticRegression(
        fit_intercept=True, l1_ratio=0.0, max_iter=1000, random_state=1234
    )
    # NOTE: For sklearn version < 1.8, use penalty='l2' instead of l1_ratio=0.0
    # lr = LogisticRegression(
    #     fit_intercept=True, penalty='l2', max_iter=1000, random_state=1234
    # )

    pipe_lr = make_pipeline(
        PolynomialFeatures(degree=degree, include_bias=False),
        StandardScaler(),
        lr,
    )
```

`validation_curve()` uses two arguments to specify which parameter should be varied:

- `param_name` specifies the attribute which stores the parameter to be varied. In case of pipelines, this name takes the form `STEPNAME__ATTRIBUTE` (note the double underscore!) where `STEPNAME` is the name of the step in the pipeline, and `ATTRIBUTE` is the name of the attribute of that particular step to be varied.
- `param_range` specifies the range of parameter values to be varied along the validation curve.

For our specific logistic regression pipeline, the call to `validation_curve()` looks as follows:

```
[35]: from sklearn.model_selection import validation_curve

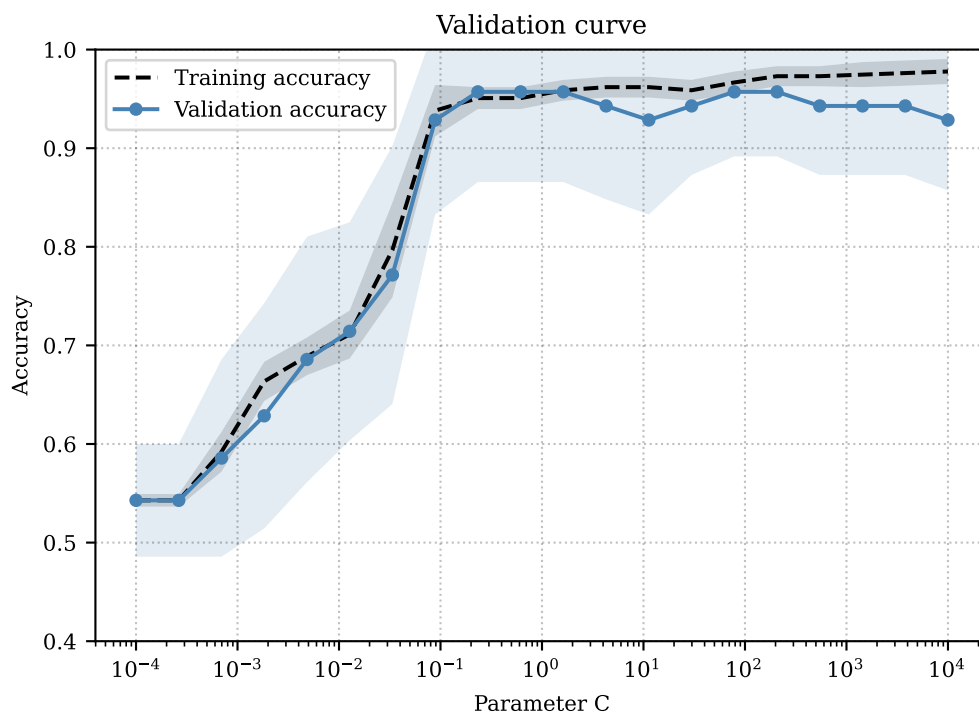
    # Define range of values for C, spaced uniformly in logs
    param_range = np.logspace(-4, 4, 20)

    # Parameter name
    param_name = 'logisticregression__C'

    # Compute validation curve
    train_scores, test_scores = validation_curve(
        estimator=pipe_lr, # Estimator pipeline
        X=X_train, # Training data
        y=y_train, # Training labels
        param_name=param_name, # Parameter name for C in pipeline
        param_range=param_range, # Range of C values to evaluate
        cv=10, # Number of folds for cross-validation
        n_jobs=4, # Number of parallel jobs
    )
```

The module `lecture4_classifiers` implements the function `plot_accuracy_validation_curve()` which we can use to plot the validation curve for a given range of parameter values:

```
[36]: from lecture4_classifiers import plot_accuracy_validation_curve
      plot_accuracy_validation_curve(param_range, train_scores, test_scores)
```



As the plot below suggests, for small C (i.e., for very large penalty strengths), the model underfits the data both in the training and test samples, leading to low accuracy. For very high values of C , on the other hand, the model overfits, leading to lower accuracy in the test sample (but increasing accuracy in the training sample).

1.2 Other commonly used classifiers

In the remainder of this lecture, we study three other widely used classifiers implemented in scikit-learn:

1. Decision trees;
2. Random forests; and
3. Support vector machines (SVM)

These classifiers should be viewed as alternatives to logistic regression studied earlier. In fact, because of the composable API of scikit-learn, it is usually trivial to switch out the `LogisticRegression` class in a processing and estimation pipeline, so one can (and should) easily experiment with several classifiers to determine which yields the best accuracy for a given task.

When running any of these classifiers, we need to be aware that they differ in terms of data processing, estimation and cross-validation:

1. Some classifiers like `LogisticRegression` have a fixed functional form (given by the sigmoid function), so we might need to expand the feature space by creating polynomial interactions or similar, as we have done previously. This step is not needed for decision trees or random forests, and it is most likely not needed for SVM because we can select a different kernel to create a nonlinear model instead of resorting to a polynomial basis.

2. Some classifiers *require* feature standardization (in particular, this is true for logistic regression with regularization and SVM), whereas this step is not necessary for decision trees or random forests.
3. Classifiers differ in their hyperparameters, and different types of cross-validation might need to be performed for different classifiers.

1.3 Decision trees

Decision trees are nonlinear classifiers that are constructed by splitting the sample in two at each branch according to some criterion (e.g., if feature 1 is larger than a value a). Fitting a decision tree is the process of growing a tree and finding the criteria that split the sample to achieve the highest information gain. In principle, a tree could be grown until each leaf node is pure, i.e., it contains only observations from a single class. In practice, this can lead to very deep trees which tend to overfit the data, so it's advisable to prune the tree by specifying a maximum tree depth.

Decision trees are implemented in the `DecisionTreeClassifier` class in scikit-learn.

To demonstrate their use, we recreate the artificial classification sample we used for logistic regression earlier.

```
[37]: # Sample size
N = 100

# Create demo data for binary classification
sigma_eps = 0.2

# Create dataset
X, y = create_class_data(N=N, sigma=sigma_eps)

# Split data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, stratify=y, test_size=0.3, random_state=1234
)
```

We grow the tree by fitting it to the training data. For the reasons explained above, we limit the maximum tree depth to 3 layers by specifying the argument `max_depth=3`.

It should be stressed that decision trees are fully nonlinear models, so there is usually no need to expand the feature space using polynomial interactions. Moreover, decision trees are not sensitive to scaling, so we can skip the entire data processing pipeline we used for other classifiers.

```
[38]: from sklearn.tree import DecisionTreeClassifier

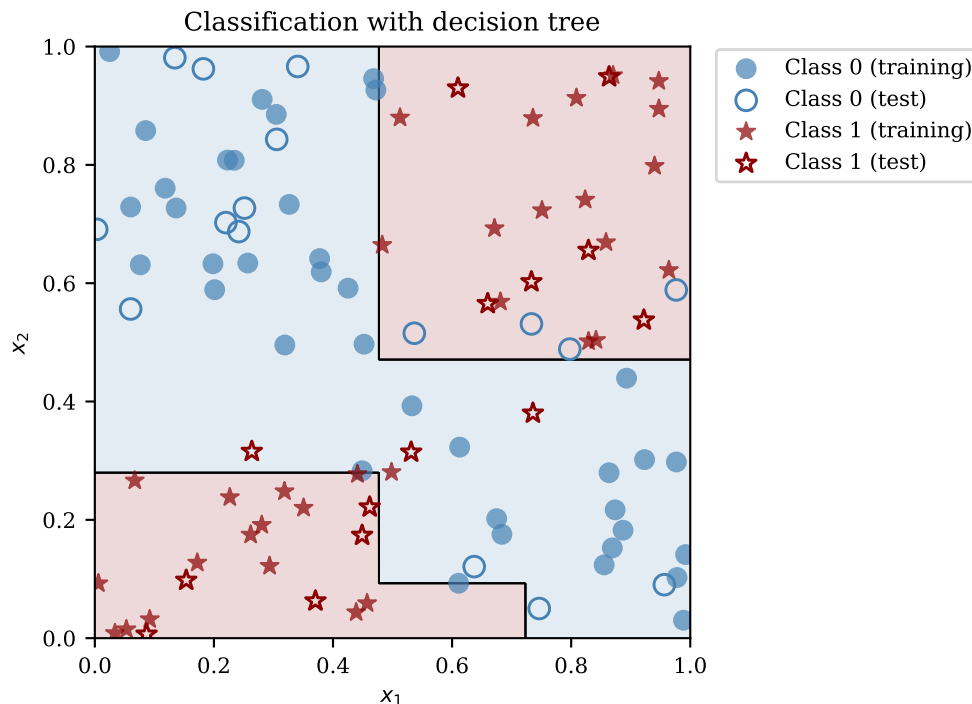
# Create decision tree classifier
tree = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=1234)

# Grow tree on training data (no transformation needed)
tree.fit(X_train, y_train)
```

```
[38]: DecisionTreeClassifier(max_depth=3, random_state=1234)
```

We now plot the decision boundaries of the fitted decision tree. As you can see, the boundaries are all parallel to the axes, which is by construction.

```
[39]: ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, tree)
_ = ax.set_title('Classification with decision tree')
```



Computing the accuracy score, we see that the decision tree produces a worse result on the test sample than the cross-validated logistic regression we used earlier:

```
[40]: y_train_pred_tree = tree.predict(X_train)
      y_test_pred_tree = tree.predict(X_test)

      acc_train = accuracy_score(y_train, y_train_pred_tree)
      acc_test = accuracy_score(y_test, y_test_pred_tree)

      print(f'Accuracy on training sample: {acc_train:.3f}')
      print(f'Accuracy on test sample: {acc_test:.3f}')
```

Accuracy on training sample: 0.986

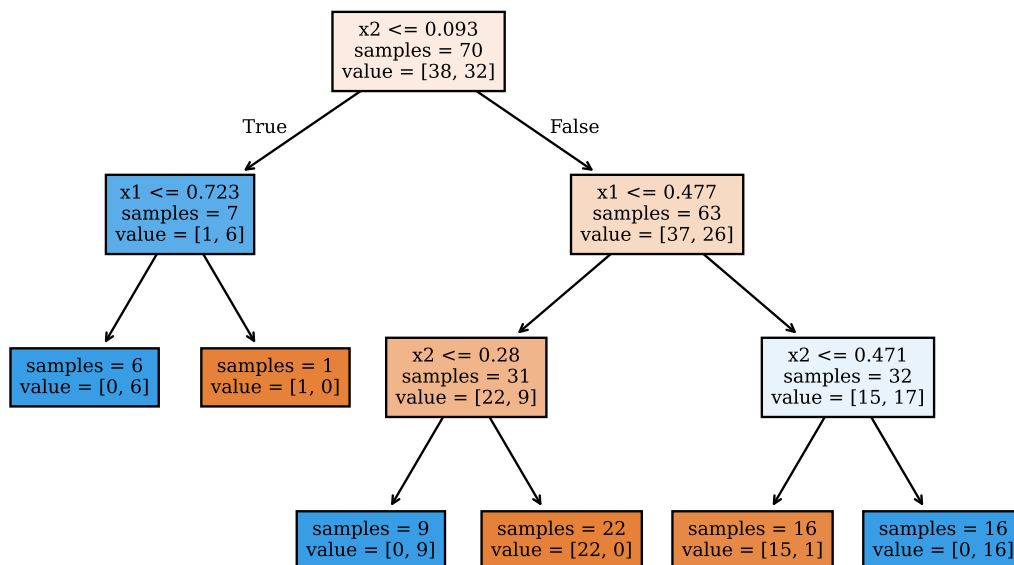
Accuracy on test sample: 0.767

scikit-learn offers a convenience function which allows us to visualize the decision tree depicted in the figure below. At each node, the first line states the criterion used to split the sample, the second line reports the sample size at the node before the split has been applied, and the third line reports the number of observations for each class label.

For example, at the top node, the sample is split according to whether the feature x_{2i} is larger than 0.093 or not. As you can see, most of the bottom leaves are pure since they contain only observations of a single class.

```
[41]: from sklearn.tree import plot_tree

      # Visualize nodes of fitted decision tree
      fig, ax = plt.subplots(figsize=(8, 5))
      _ = plot_tree(
          tree, filled=True, feature_names=['x1', 'x2'], fontsize=8, impurity=False, ax=ax
      )
```



We fitted the tree above choosing a maximum tree depth ourselves. Alternatively, we can use the `GridSearchCV` cross-validator to find the optimal tree depth for this sample. The parameter grid is specified using the `param_grid` argument which accepts a dictionary with a list of candidate values for *each* parameter (in this case only `max_depth`, but we'll add more parameters to cross-validate in later sections).

```
[42]: from sklearn.model_selection import GridSearchCV

# Define range of values for max_depth
param_range = np.arange(1, 11)

tree_cv = GridSearchCV(
    estimator=tree, param_grid={'max_depth': param_range}, cv=10, refit=True, n_jobs=4
).fit(X_train, y_train)

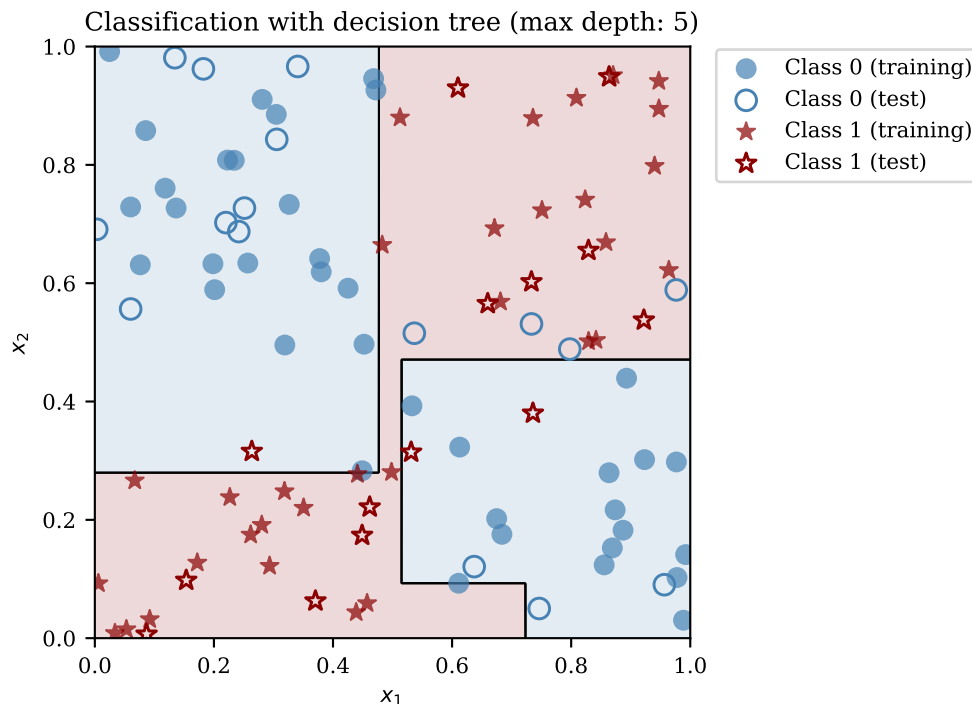
print(f'Best accuracy: {tree_cv.best_score_:.3f}')
print(f'Best parameters: {tree_cv.best_params_}')
```

Best accuracy: 0.886

Best parameters: {'max_depth': np.int64(5)}

As you can see, the optimal cross-validated maximum depth is computed to be 5.

```
[43]: ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, tree_cv)
_ = ax.set_title(
    f'Classification with decision tree (max depth: {tree_cv.best_params_["max_depth"]})'
)
```



The cross-validated maximum depth perfectly classifies all observations in the training sample, but does not increase accuracy on the test sample compared to the tree we grew earlier with `max_depth=3`.

```
[44]: y_train_pred_tree = tree_cv.predict(X_train)
      y_test_pred_tree = tree_cv.predict(X_test)

      acc_train = accuracy_score(y_train, y_train_pred_tree)
      acc_test = accuracy_score(y_test, y_test_pred_tree)

      print(f'Accuracy on training sample: {acc_train:.3f}')
      print(f'Accuracy on test sample: {acc_test:.3f}')
```

```
Accuracy on training sample: 1.000
Accuracy on test sample: 0.767
```

1.4 Random forest

A random forest classifier combines multiple decision trees on various subsamples of the data and aggregates their results by majority vote. This model averaging is intended to lead to less overfitting and higher predictive accuracy.

The random forest classifier is implemented in `RandomForestClassifier` in `scikit-learn`. The most important arguments are the number of decision trees to grow (`n_estimators`) and the maximum tree depth (`max_depth`).

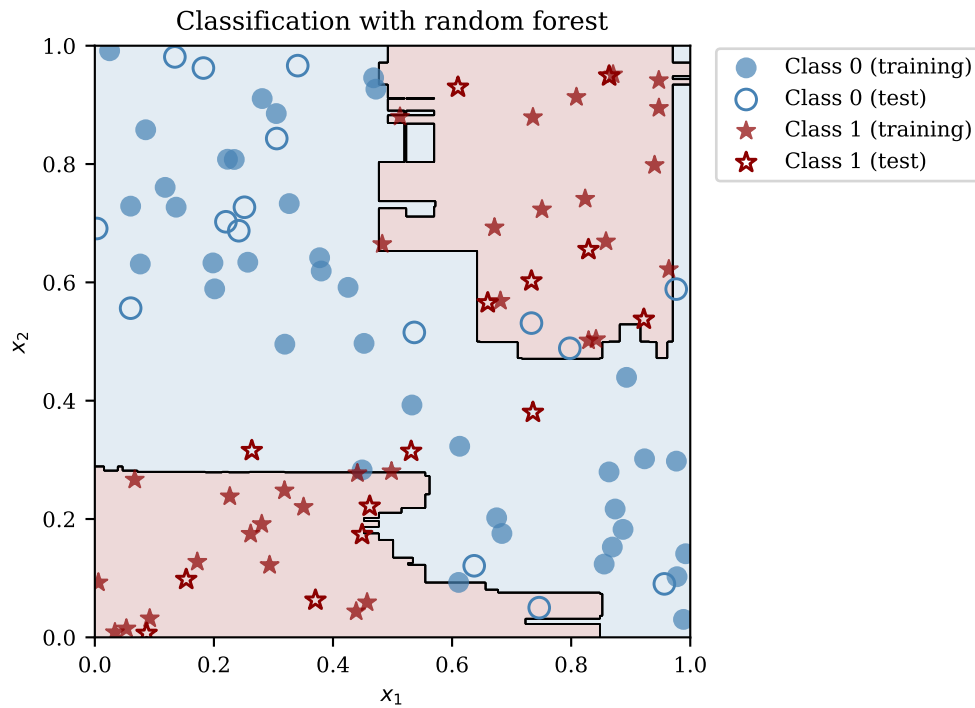
Just like decision trees, a random forest is not sensitive to scaling, so we can skip the usual preprocessing pipeline.

```
[45]: from sklearn.ensemble import RandomForestClassifier

      # Create random forest classifier, fit on training data
      forest = RandomForestClassifier(
          n_estimators=100, max_depth=3, random_state=1234, n_jobs=4
      ).fit(X_train, y_train)
```

Compared to a single decision tree, the random forest leads to more granular decision boundaries which are due to the averaging of many decision trees.

```
[46]: ax = plot_classes(X_train, y_train, X_test, y_test)
      plot_decision_boundary(ax, xvalues, forest)
      _ = ax.set_title('Classification with random forest')
```



As you can see, the random forest already has a higher prediction accuracy than a single tree, even without cross-validation.

```
[47]: y_train_pred_forest = forest.predict(X_train)
      y_test_pred_forest = forest.predict(X_test)

      acc_train = accuracy_score(y_train, y_train_pred_forest)
      acc_test = accuracy_score(y_test, y_test_pred_forest)

      print(f'Accuracy on training sample: {acc_train:.3f}')
      print(f'Accuracy on test sample: {acc_test:.3f}')
```

Accuracy on training sample: 0.986

Accuracy on test sample: 0.800

There is no dedicated cross-validation class for random forest, but we can use [GridSearchCV](#) to determine the maximum tree depth (`max_depth`) or the number of trees to grow (`n_estimators`) in the random forest. The grid search is then performed over the Cartesian product of these candidate values, in this case of 10×10 different parameter combinations. Other than that, this works analogously to what we did for decision trees:

```
[48]: from sklearn.model_selection import GridSearchCV

      # Define range of values for max_depth
      max_depth_grid = np.arange(1, 11)

      # Define range of values for n_estimators
      n_estimators_grid = np.arange(100, 200, 10)

      # Define dictionary of hyperparameters to search over
```

```

param_grid = {'max_depth': max_depth_grid, 'n_estimators': n_estimators_grid}

# Run grid search over two hyperparameters
forest_cv = GridSearchCV(
    estimator=forest, # Estimator to use
    param_grid=param_grid, # Dictionary of hyperparameters to search over
    cv=10, # Number of folds for cross-validation
    refit=True, # Refit model with best hyperparameters on entire training data
    n_jobs=8, # Number of parallel jobs
).fit(X_train, y_train)

print(f'Best accuracy: {forest_cv.best_score_:.3f}')
print(f'Best parameters: {forest_cv.best_params_}')

```

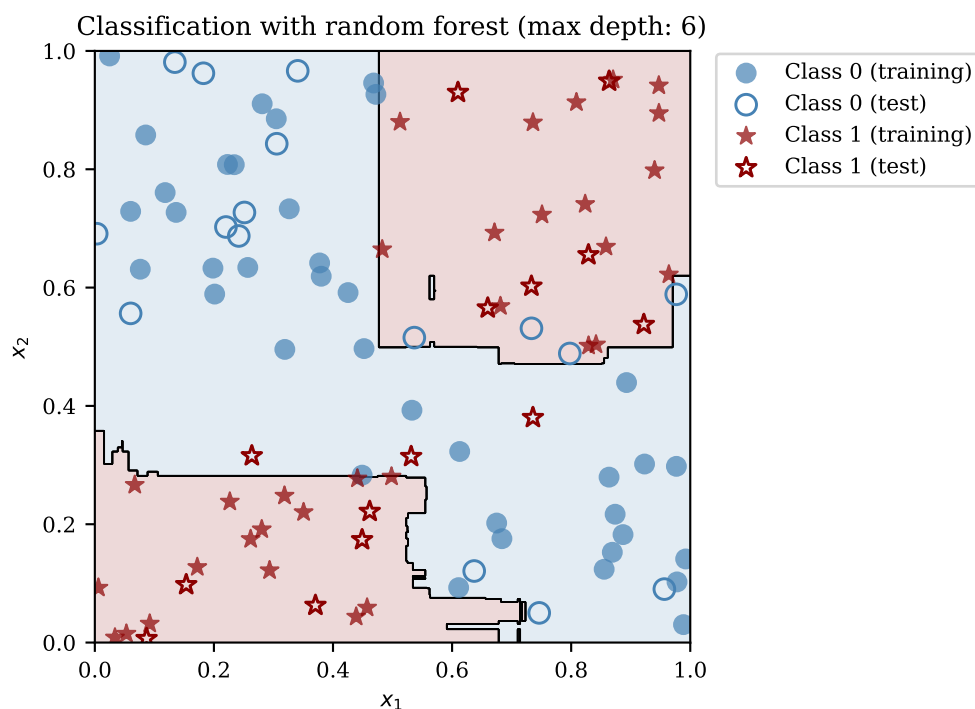
Best accuracy: 0.929

Best parameters: {'max_depth': np.int64(6), 'n_estimators': np.int64(130)}

```

[49]: ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, forest_cv)
_ = ax.set_title(
    f'Classification with random forest (max depth: {forest_cv.best_params_["max_depth"]})'
)

```



As the output below shows, the cross-validated maximum tree depth of 6 does lead to an improvement in accuracy on the training sample, but not to additional improvements on the test sample:

```

[50]: y_train_pred_forest = forest_cv.predict(X_train)
y_test_pred_forest = forest_cv.predict(X_test)

acc_train = accuracy_score(y_train, y_train_pred_forest)
acc_test = accuracy_score(y_test, y_test_pred_forest)

print(f'Accuracy on training sample: {acc_train:.3f}')
print(f'Accuracy on test sample: {acc_test:.3f}')

```

Accuracy on training sample: 1.000

Accuracy on test sample: 0.800

1.5 Support vector machines (SVM)

scikit-learn offers several implementations of SVMs that differ in the underlying algorithm and capabilities. We focus on the [SVC](#) (support vector classifier) class which supports both linear and nonlinear kernels.

Just like logistic regression, SVC uses the regularization parameter C . However, SVMs offer additional flexibility in that they support various kernels. You can think of a kernel as a function that operates on the features and projects them onto a different space. For example, we might have a classification problem like the one we studied above, where the classes 0 and 1 clearly cannot be separated by a single linear boundary. Instead of adding polynomial interactions, with SVMs we can specify a nonlinear kernel in the hope that once the features have been transformed using this kernel, the individual classes become much easier to separate.

1.5.1 Linear kernel

For illustration purposes, we first create a new artificial dataset for classification with more observations and more noise to magnify the difference between SVM kernels and hyperparameters (with the original demo sample, all but the simplest linear SVM yield basically the same result).

```
[51]: # Sample size
      N = 200

      # Create demo data for binary classification
      sigma_eps = 0.25

      # Create dataset
      X, y = create_class_data(N=N, sigma=sigma_eps)

      # Split data into training and test sets
      X_train, X_test, y_train, y_test = train_test_split(
          X, y, stratify=y, test_size=0.3, random_state=1234
      )
```

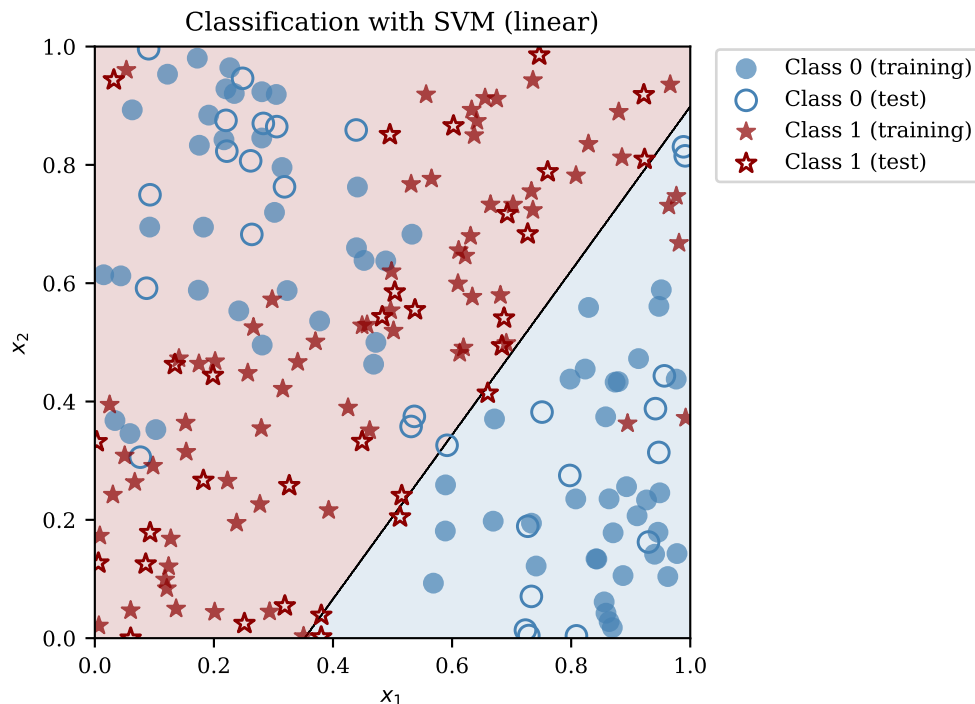
We start with the simplest linear kernel which is selected using the `kernel='linear'` argument.

```
[52]: from sklearn.pipeline import make_pipeline
      from sklearn.preprocessing import PolynomialFeatures, StandardScaler
      from sklearn.svm import SVC

      # Create pipeline with linear SVM
      pipe_svm = make_pipeline(
          StandardScaler(), SVC(kernel='linear', C=1.0, random_state=1234)
      ).fit(X_train, y_train)
```

Plotting the decision boundaries, we see that the linear kernel leads to a single linear decision boundary:

```
[53]: xvalues = np.linspace(0, 1, 1000)
      ax = plot_classes(X_train, y_train, X_test, y_test)
      plot_decision_boundary(ax, xvalues, pipe_svm)
      _ = ax.set_title('Classification with SVM (linear)')
```



This model does not perform particularly well in terms of accuracy, classifying only about 70% of test samples correctly:

```
[54]: y_train_pred_svm = pipe_svm.predict(X_train)
y_test_pred_svm = pipe_svm.predict(X_test)

acc_train = accuracy_score(y_train, y_train_pred_svm)
acc_test = accuracy_score(y_test, y_test_pred_svm)

print(f'Accuracy on training sample: {acc_train:.3f}')
print(f'Accuracy on test sample: {acc_test:.3f}')
```

Accuracy on training sample: 0.729

Accuracy on test sample: 0.683

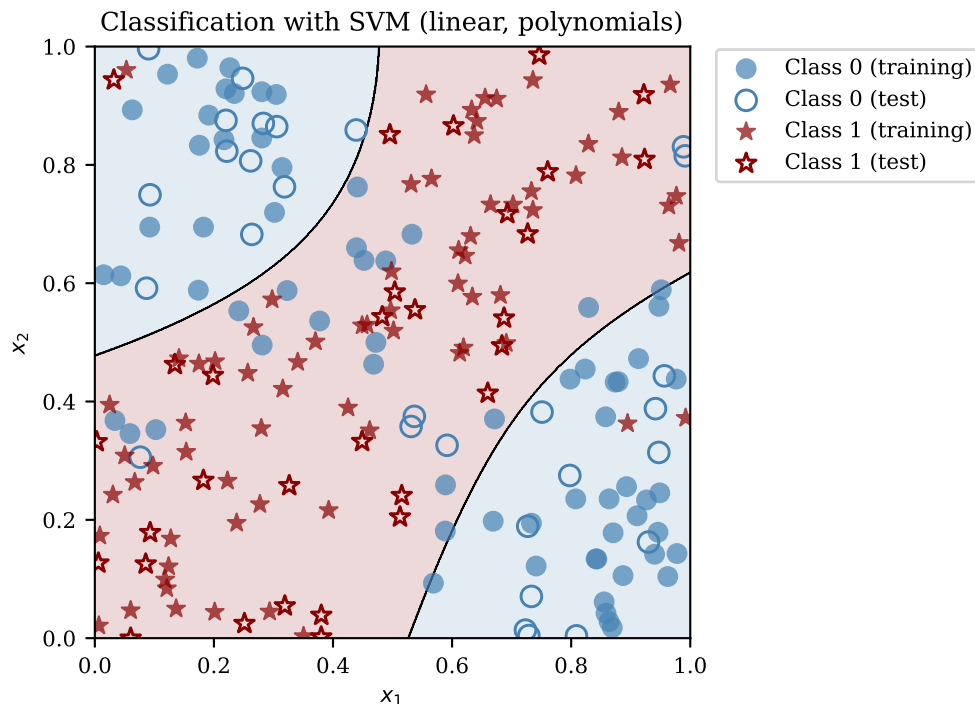
As with logistic regression, we can easily augment this model by adding polynomial interactions of features:

```
[55]: # Fit linear SVM with polynomial features

# Maximum polynomial degree
degree = 5

# Create pipeline with polynomial features and SVM
pipe_svm_poly = make_pipeline(
    PolynomialFeatures(degree=degree, include_bias=False),
    StandardScaler(),
    SVC(kernel='linear', C=1.0, random_state=1234),
).fit(X_train, y_train)
```

```
[56]: xvalues = np.linspace(0, 1, 1000)
ax = plot_classes(X_train, y_train, X_test, y_test)
plot_decision_boundary(ax, xvalues, pipe_svm_poly)
_ = ax.set_title('Classification with SVM (linear, polynomials)')
```



This yields considerably higher accuracy on the test sample:

```
[57]: y_train_pred_svm = pipe_svm_poly.predict(X_train)
      y_test_pred_svm = pipe_svm_poly.predict(X_test)

      acc_train = accuracy_score(y_train, y_train_pred_svm)
      acc_test = accuracy_score(y_test, y_test_pred_svm)

      print(f'Accuracy on training sample: {acc_train:.3f}')
      print(f'Accuracy on test sample: {acc_test:.3f}')
```

Accuracy on training sample: 0.850

Accuracy on test sample: 0.883

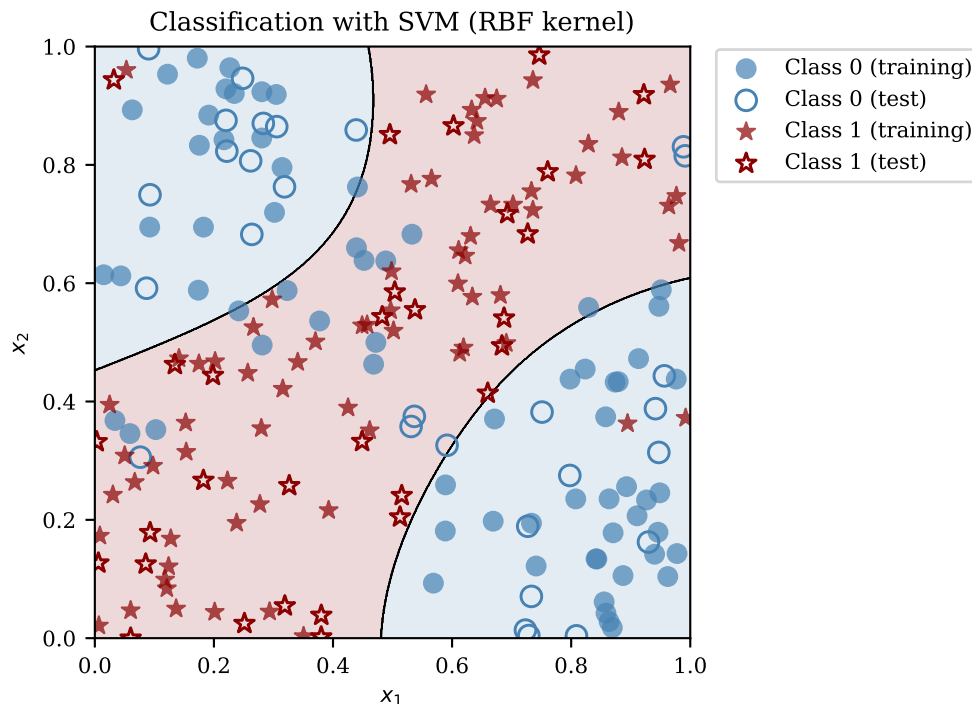
1.5.2 Gaussian kernel

However, it is not necessary to introduce polynomial features with SVM. Alternatively, we can operate with just the two original features, but choose a different kernel. By default, the kernel used by SVC is `kernel='rbf'` which stands for the *radial basis function*. This kernel is also called the Gaussian kernel.

We rerun the above code, omitting the polynomial features but instead selecting the Gaussian kernel:

```
[58]: # Create SVM with RBF kernel
      pipe_svm_rbf = make_pipeline(
          StandardScaler(), SVC(kernel='rbf', C=1.0, gamma='scale', random_state=1234)
      ).fit(X_train, y_train)
```

```
[59]: ax = plot_classes(X_train, y_train, X_test, y_test)
      plot_decision_boundary(ax, xvalues, pipe_svm_rbf)
      _ = ax.set_title('Classification with SVM (RBF kernel)')
```



The decision boundaries are quite similar to the linear kernel with polynomials, as is the accuracy of the fitted model:

```
[60]: y_train_pred_svm = pipe_svm_rbf.predict(X_train)
y_test_pred_svm = pipe_svm_rbf.predict(X_test)

acc_train = accuracy_score(y_train, y_train_pred_svm)
acc_test = accuracy_score(y_test, y_test_pred_svm)

print(f'Accuracy on training sample: {acc_train:.3f}')
print(f'Accuracy on test sample: {acc_test:.3f}')
```

Accuracy on training sample: 0.879

Accuracy on test sample: 0.883

So far, we haven't done any hyperparameter tuning. The RBF kernel has *two* hyperparameter arguments:

1. C , the regularization penalty strength (which we encountered before); and
2. γ , which scales the squared Euclidean distance between any two feature vectors.

With two hyperparameters, we cannot resort to functions such as the `validation_curve()` we used earlier to plot how accuracy changes with a hyperparameter. Instead, we use the `GridSearchCV` class which performs parameter tuning over a grid of *several* parameters (in this case, over C and γ). The parameter grid is specified using the `param_grid` argument which accepts a dictionary with a list of candidate values for *each* parameter. The grid search is then performed over the Cartesian product of these candidate values, so in the code below, 50×50 different combinations of parameters are evaluated.

```
[61]: from sklearn.model_selection import GridSearchCV

# Define range of values for C and gamma, spaced uniformly in logs
param_range = np.logspace(-1, 2, 50)

# Define param_grid argument for GridSearchCV
param_grid = {'svc__C': param_range, 'svc__gamma': param_range}
```

```
# Create grid search cross-validation object, run cross-validation
gs_rbf = GridSearchCV(
    estimator=pipe_svm_rbf, # Model to tune
    param_grid=param_grid, # Dictionary of hyperparameter grids
    cv=5, # Number of folds
    refit=True, # Refit best model
    n_jobs=4, # Number of parallel jobs
).fit(X_train, y_train)
```

The best accuracy score and the optimal parameter values can be retrieved from the attributes `best_score_` and `best_params_`:

```
[62]: print(f'Best accuracy: {gs_rbf.best_score_:.3f}')
      print(f'Best parameters: {gs_rbf.best_params_}')
```

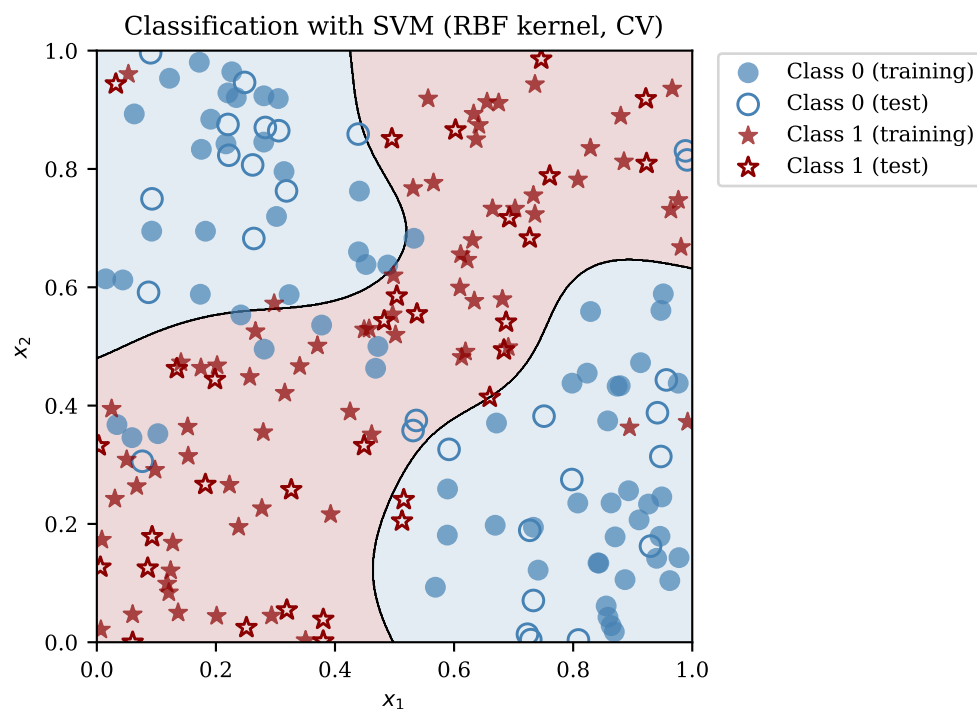
Best accuracy: 0.900

Best parameters: {'svc__C': np.float64(0.47148663634573934), 'svc__gamma': np.float64(3.906939937054617)}

Note that we don't need to re-estimate the model with these optimal parameters since we specified `refit=True` when creating an instance of `GridSearchCV`, so the resulting object already reflects the optimal model.

Finally, we can plot the decision boundaries for the cross-validated SVM model:

```
[63]: ax = plot_classes(X_train, y_train, X_test, y_test)
      plot_decision_boundary(ax, xvalues, gs_rbf)
      _ = ax.set_title('Classification with SVM (RBF kernel, CV)')
```



As you might suspect from the figure, the cross-validated model is in fact overfitting more than the one with default hyperparameters. This is confirmed by the accuracy score, which is *lower* on the test sample than it was before.

```
[64]: y_train_pred_svm = gs_rbf.predict(X_train)
      y_test_pred_svm = gs_rbf.predict(X_test)
```

```
acc_train = accuracy_score(y_train, y_train_pred_svm)
acc_test = accuracy_score(y_test, y_test_pred_svm)

print(f'Accuracy on training sample: {acc_train:.3f}')
print(f'Accuracy on test sample: {acc_test:.3f}')
```

Accuracy on training sample: 0.907

Accuracy on test sample: 0.850

1.6 List of functions used in this lecture

The following list contains the central functions covered in this lecture. Each function name is linked to the official API documentation which you can consult for more details regarding function arguments and return values. The [\[go to section\]](#) links to the relevant section in this notebook.

1.6.1 NumPy array manipulation functions

- [meshgrid\(\)](#): Function to create coordinate matrices from coordinate vectors. [\[go to section\]](#)
- [column_stack\(\)](#): Function to stack 1-D arrays as columns into a 2-D array. [\[go to section\]](#)

1.6.2 SciPy statistics functions

- [logistic.cdf\(\)](#): Logistic cumulative distribution function. [\[go to section\]](#)

1.6.3 Scikit-learn classification models

- [LogisticRegression\(\)](#): Class for fitting logistic regression models. [\[go to section\]](#)
- [LogisticRegressionCV\(\)](#): Class for fitting logistic regression models with built-in cross-validation. [\[go to section\]](#)
- [DecisionTreeClassifier\(\)](#): Class for fitting decision tree classifiers. [\[go to section\]](#)
- [RandomForestClassifier\(\)](#): Class for fitting random forest classifiers. [\[go to section\]](#)
- [SVC\(\)](#): Class for fitting support vector machine classifiers. [\[go to section\]](#)

1.6.4 Scikit-learn metrics and visualization

- [accuracy_score\(\)](#): Function for computing accuracy classification score. [\[go to section\]](#)
- [precision_score\(\)](#): Function for computing precision classification score. [\[go to section\]](#)
- [recall_score\(\)](#): Function for computing recall classification score. [\[go to section\]](#)
- [f1_score\(\)](#): Function for computing F1 classification score. [\[go to section\]](#)
- [ConfusionMatrixDisplay](#): Class for visualizing confusion matrices. [\[go to section\]](#)
- [plot_tree\(\)](#): Function for visualizing decision trees. [\[go to section\]](#)

1.6.5 Scikit-learn model selection and evaluation

- [GridSearchCV\(\)](#): Class for performing grid search with cross-validation. [\[go to section\]](#)
- [validation_curve\(\)](#): Function for computing validation curves. [\[go to section\]](#)